Operating Systems: Assignment Submission

MEMORY MANAGEMENT & MMU HARDWARE ARCHITECTURE

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LOGICAL ADDRESS	RELOCATION REGISTER	LIMIT REGISTER	STATUS
346	14,000	10,500	Valid / Verified

1. Physical Address Calculation Sheet

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Physical Address = Logical Address + Relocation Register
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Step A: Address Fetch & Range Validation
The CPU issues a logical request for address **346**. The MMU validates that 0 ≤ 346 < Limit (10,500), confirming access is secure.

Step B: Base Offset Addition
The hardware adder calculates the physical offset by summing the logical address with the process's relocation base: **346 + 14,000 = 14,346**.

CALCULATED FINAL PHYSICAL ADDRESS 14,346

2. Protection Check Logic Flowchart (protection_check.png)

1. INPUT CAPTURE
MMU receives Logical Address (LA = 346) and Limit Register (L = 10,500).

2. BOUNDARY CHECK
Evaluates 0 ≤ LA < L. Since 346 < 10,500, condition passes successfully.

3. FAULT HANDLING
If LA ≥ L, immediately trigger Addressing Error Trap (Segmentation Fault).

3. MMU Role Summary Report (memory_unit_role.txt)

The Memory Management Unit (MMU) is a hardware component responsible for real-time translation between logical addresses generated by the CPU and physical addresses in main memory. During execution, the MMU intercepts every memory reference. It first performs a protection check by comparing the incoming logical address against the limit register to ensure the access is within the process's allocated bounds, preventing unauthorized memory access or segmentation faults. If valid, the MMU adds the value of the relocation register to the logical address, producing the final physical address sent to memory. This hardware-level mechanism ensures process isolation, memory protection, and seamless relocation without software overhead.